THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In photosynthesis, the oxygen released comes from: 1

(a) glucose **(b)** carbon dioxide **(c)** chlorophyll **(d)** water

2. In a reflex arc, the correct order is: 1

(a) receptor -> sensory neuron -> spinal cord -> motor neuron -> effector

(b) effector -> motor neuron -> spinal cord -> sensory neuron -> receptor

(c) receptor -> motor neuron -> spinal cord -> sensory neuron -> effector

(d) receptor -> spinal cord -> brain -> effector

3. In which of the following does the offspring differ least from the parent? 1

(a) Offspring produced by seeds **(b)** Offspring produced by fertilisation

(c) Offspring produced by budding in Hydra **(d)** Offspring produced by cross-pollination

4. A heterozygous tall plant (Tt) is crossed with a dwarf plant (tt). The fraction of dwarf plants in the offspring is 1

(a) 0 **(b)** 1/2 **(c)** 3/4 **(d)** 1/4

5. Which has the longest small intestine? 1

(a) humans **(b)** birds

(c) carnivores like tiger **(d)** herbivores such as the cow

6. Why are shoots positively phototropic and roots negatively phototropic? 1

(a) roots have no auxin

(b) shoots need gravity

(c) auxin affects stem and root cells in opposite ways

(d) roots need light

7. The thread-like structure on which anther is held is the 1

(a) ovule **(b)** style **(c)** filament **(d)** stigma

8. The sex of a baby is determined at the time of 1

(a) puberty **(b)** implantation **(c)** fertilisation **(d)** birth

9. The blood vessel carrying deoxygenated blood from the heart to the lungs is: 1

(a) pulmonary vein **(b)** vena cava **(c)** aorta **(d)** pulmonary artery

10. A person has frequent thirst and urination and sugar in urine. Which hormone is deficient? 1

(a) growth hormone **(b)** thyroxine **(c)** adrenaline **(d)** insulin

11. How many chromosomes would a human egg cell have, if the body cell has 46? 1

(a) 92 **(b)** 23 **(c)** 46 **(d)** 12

12. A trait that is controlled by a recessive allele appears in a plant when 1

(a) both alleles are recessive **(b)** one allele is recessive

(c) both are dominant **(d)** no allele is present

13. Which of the following is NOT a requirement for photosynthesis? 1

(a) sunlight **(b)** water **(c)** chlorophyll **(d)** oxygen

14. The connection between sensory and motor neurons inside the spinal cord is made by: 1

(a) dendrite **(b)** axon **(c)** effector **(d)** relay neuron

15. A unicellular organism that divides into many cells inside a protective cyst is 1

(a) Yeast **(b)** Paramecium **(c)** Amoeba **(d)** Plasmodium

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a leaf kept in dark not show starch? How is this tested? 2

17. Why are reflexes important for survival? 2

18. Why do single-celled organisms reproduce by fission while many-celled complex organisms do not? 2

19. In pea plants, tall (T) is dominant. A tall plant is crossed with a dwarf plant and 50% dwarf offspring result. Find the genotype of the tall parent. 2

20. Why do herbivores have a longer small intestine than carnivores? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how the leaf is adapted for photosynthesis. 3

22. A person's cerebrum is damaged. Which functions are affected? 3

23. Explain why a large number of spores are produced by Rhizopus even though only a few land in a suitable place. 3

24. A tall pea plant is crossed with a dwarf and the F1 are all tall. When the F1 are self-pollinated, 300 plants result. Find the numbers of tall and dwarf plants and the genotypes. 3

OR

Explain why the oxygenated and deoxygenated blood must not mix in the heart. 3

25. Describe the role of feedback in the control of thyroxine. 3

OR

Why is there a wide difference in the number of male and female gametes in many organisms? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe an experiment to show that sunlight is necessary for photosynthesis or that CO_2 is needed, with the observations. 5

OR

Trace the path of an impulse when you step on a thorn: describe receptors, neurons, brain, response and why the foot moves before pain is felt. 5

27. Describe different methods of asexual reproduction in unicellular and multicellular organisms. State why they are not used by complex animals. 5

OR

In pea plants, round seed (R) is dominant to wrinkled (r), and yellow (Y) to green (y). A plant RrYy is self-pollinated. Make a Punnett square outline: list gametes, give the F2 phenotypes and the ratio. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND